

WEERAYUT BUAPHET

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SUMMARY

AI engineer and NLP researcher specializing in information extraction, LLMs, Retrieval-Augmented Generation (RAG), tool-use agents, and multilingual NLP. Experienced in developing medical AI systems and conducting research on low-resource NLP, few-shot learning, and text normalization.

Currently developing tool-use AI agents for automated ICD-10 clinical coding in hospital ASR workflows at an AI health-tech startup. Also contributing to the [ThaiLLM](#) project through dataset creation, model development, and evaluation for a factually grounded multi-turn question-answering system, with a focus on retrieval optimization and LLM post-training methods, including supervised fine-tuning and preference optimization.

EDUCATION

Ph.D. in Information Science and Technology (5-year program, fully funded) Aug 2020 – Mar 2026
Vidyasirimedhi Institute of Science and Technology (VISTEC) 4.00/4.00 GPA
Relevant coursework: Natural Language Processing, Computational Machine Intelligence and Applications

B.Eng. in Computer Engineering Mar 2016 - Mar 2020
Rajamangala University of Technology Lanna, Chiang Mai. 3.62/4.00 GPA (Ranked 1st out of 116)
Relevant coursework: Data Structures and Algorithms, Operating Systems, Software Engineering

WORK EXPERIENCE & INTERNSHIPS

Cariva (AI health-tech startup), Bangkok, Thailand: AI Engineer Mar 2026 – Present

- Designed and implemented a tool-use AI agent framework for automated ICD-10 clinical coding, integrated into ASR systems for hospitals.
- Developed prompting strategies and analyzed pipeline failure modes to handle system prompt uncertainty, collaborating with MD teams to propose and validate improvements.

VISTEC, Rayong, Thailand: Research Assistant Sep 2025 – Feb 2026 (6 months)

- Developed a RAG-based chatbot for multi-turn medical QA as part of the ThaiLLM project (Big Data Institute). Optimized retrieval using a vector database with LlamaIndex, designed prompting strategies, and improved model performance via supervised fine-tuning (SFT) and preference optimization (DPO, GRPO).

ITU, Copenhagen, Denmark: PhD Internship (Supervisor: Assoc. Prof. Rob van der Goot) Sep 2024 – Jul 2025

- Conducted research on few-shot Named Entity Recognition, evaluating encoder-based and LLM-based models and analyzing their trade-offs in few-shot settings.
- Conducted research on multi-lexical normalization for Asian languages, collaborating with five international teams from Japan, Korea, Vietnam, Indonesia, and Thailand to create a multilingual Asian text normalization benchmark and introduce LLM-based text normalization methods.
- Implemented tokenizers (morpheme, unigram, and bigram) to support Small Language Model (SLM) experiments, studying the impact of tokenizations on SLM and downstream tasks.

VISTEC, Rayong, Thailand: NLP Research Projects Nov 2019 – Jul 2024

- Led end-to-end annotation pipelines for two NER datasets used in text anonymization and downstream NLP tasks: the SCB Thai NNER dataset (outsourced annotators) and a bilingual financial NER dataset funded by the Capital Market Development Fund (internal annotation). Responsibilities included designing entity schema and annotation guidelines, data preparation, quality control, annotator coordination, and baseline model development.

TECHNICAL SKILLS

- Languages:** Thai (Native), English (B2-level in 2024)
- Domains:** Clinical NLP, RAG Systems, Few-Shot Learning, Information Extraction, Text Normalization
- Tools:** Python, PyTorch, SLURM, Docker, Tool-Calling AI Agents, LangChain, FastAPI, Streamlit

ACADEMIC PROJECTS

- **Thai Nested Named Entity Recognition Corpus (Thai N-NER).** May 2022
Weerayut Buaphet, Can Udomcharoenchaikit, Peerat Limkonchotiwat, Attapol Rutherford, and Sarana Nutanong. 2022. In Findings of the Association for Computational Linguistics: ACL 2022, pages 1473–1486, Dublin, Ireland. Association for Computational Linguistics.
- **LLM-Augmented Prototype Representation for Few-Shot Named-Entity Recognition** Nov 2025
Weerayut Buaphet, Peerat Limkonchotiwat, Attapol Rutherford, Can Udomcharoenchaikit, and Sarana Nutanong. 2025. IEEE Access.
- **MultiLexNorm++: A Unified Benchmark and a Generative Model for Lexical Normalization for Asian Languages** Accepted to TALLIP (in production)
Weerayut Buaphet, Thanh-Nhi Nguyen, Risa Kondo, Tomoyuki Kajiwar, ..., Rob van der Goot. 2025. ACM Transactions on Asian and Low-Resource Language Information Processing.
- **Cross-Lingual Data Augmentation for Thai Question-Answering.** Dec 2023
Parinthapat Pengpun, Can Udomcharoenchaikit, Weerayut Buaphet, and Peerat Limkonchotiwat. 2023. In Proceedings of the 1st GenBench Workshop on (Benchmarking) Generalisation in NLP, pages 193–203, Singapore. Association for Computational Linguistics.
- **Mitigating Spurious Correlation in Natural Language Understanding with Counterfactual Inference** Dec 2022
Can Udomcharoenchaikit, Wuttikorn Ponwitararat, Patomporn Payoungkhamdee, Kanruethai Masuk, Weerayut Buaphet, Ekapol Chuangsuwanich, and Sarana Nutanong. 2022. In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, pages 11308–11321, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- **Bilingual Named Entity Recognition for Finance (Fin-NER).** Under review
Creating a finance-NER dataset composed of two languages, Thai and English. This project aims to study the knowledge transfer from high-resource to low-resource languages in the financial domain.

CONTRIBUTIONS AND ACTIVITIES

- **Reviewer and Organizer**
Reviewer for ARR-EMNLP 2024–2026 and ACL 2026.
Co-organizer of [W-NUT 2025](#) (NAACL 2025), [W-NUT 2026](#), and [MultiLexNorm 2026](#) (EMNLP 2026).
- **Mentor AI Builders (2022)**
Co-mentored five student projects on question generation, fake news detection, and plant tissue laboratory dataset/system development. One project won the Best Presentation Award and was published at the GenBench Workshop at ACL 2023
- **Internet of Things (11th 9 RMUT competition 2019) - RMUTSB**
We got 2nd place in the RMUT group IoT competition in Thailand. We used an ESP20 to read sensor data and send it via MQTT to a Raspberry Pi server, which visualized the data on a web interface. I programmed the visualization and configured the ESP20, while my teammate handled the hardware connections.
- **The Robotic Design Contest (RDC 2018)**
This program selects national representatives for the International Design Contest RoBoCon (IDC RoBoCon). All teams are required to design a robot to solve a provided problem. It promotes equality with mixed teams, equal resources, and collaboration, including training sessions at all levels. My teammate and I achieved the following:
 - 2nd place in the Northern region at Chiang Mai University.
 - 1st place in Thailand at Chulalongkorn University.
 - [3rd place internationally at the Tokyo Institute of Technology.](#)